ELEMENTOPIA

Elementopia is an interactive, web-based gamified learning platform that transforms complex chemistry concepts into engaging adventures for high school students.

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Traditional computer-assisted chemistry simulations frequently isolate abstract atomic principles from exploratory practice, encouraging passive memorization or riddle-style shortcuts over behavioral problem-solving. This paper details the design and development of *Elementopia*, a secure, web-based gamified learning platform engineered to improve conceptual chemistry mastery among high school students. While an early prototype established basic bonding canvas interactions, it remained highly vulnerable to client-side state manipulation and lacked structural progression constraints. To bridge these architectural gaps, we present a complete engineering overhaul transforming the system into a server-authoritative ecosystem governed by a Spring Boot REST API backend, a modular React.js frontend tier, and a cloud-provisioned Supabase PostgreSQL database layer.

The system implements a server-authoritative validation service that intercepts structural element submissions, cross-references configurations against database dictionaries, and returns custom diagnostic byproduct micro-lessons to correct underlying misconceptions in real-time. To eliminate rapid-guessing exploits, an active temporal security engine enforces a sliding-window threshold that triggers a temporary client-side interface freeze if a user logs five consecutive failures within fifteen seconds. Stateful access control gatekeeping locks advanced domains until a session verifies a threshold of three correct reactions, while an algorithmic challenge randomizer shuffles remaining tasks by filtering out a student's past discoveries. Developed via an Agile software engineering methodology, the resulting architecture removes local data tampering vectors and optimizes data logging pipelines for high-concurrency classroom utilization.

CCS CONCEPTS: • Applied computing → Education • Applied computing → Interactive learning environments • Human-centered computing → Web-based interaction • Software and its engineering → Web applications • Applied computing → Computer-assisted instruction

Additional Keywords and Phrases: Gamification, Spring Boot, Server-Authoritative State, Sliding Window Algorithm, Telemetry Data Pipeline, Chemistry Education, Relational Database Design

1 Introduction

Chemistry has long been recognized as a foundational yet notoriously difficult pillar of secondary science education [1]. The primary teaching obstacle stems from the inherently abstract and multi-level nature of

chemical phenomena, which require students to seamlessly translate between macroscopic visual changes, microscopic molecular structures, and symbolic representations. In traditional educational environments, core concepts such as atomic arrangements, electronic configurations, and simulated bonding mechanics are frequently taught through static blackboard illustrations or verbal explanations [4]. This reliance on passive memorization heavily limits conceptual understanding and leaves students with persistent knowledge gaps [1]. So without dynamic, goal-directed interactions, learners struggle to visualize how micro-level subatomic changes dictate macro-level chemical reactivity, leading to disengagement and diminished learning outcomes [7].

Now to quantify these localized educational struggles, a foundational diagnostic survey was conducted among twenty-five (25) high school chemistry students. The empirical results explicitly confirmed severe motivational and cognitive deficits in the current learning ecosystem: seventy-two percent (72%) of surveyed students stated that they find it exceptionally difficult to memorize and comprehend chemical elements and their subatomic relationships using traditional materials, while ninety-six percent (96%) explicitly expressed a preference for learning within an interactive, gamified digital framework. Educational literature strongly reinforces this directional shift, confirming that structurally coherent gamified environments promote optimal cognitive processing, reduce extraneous cognitive load, and transform passive instruction into meaningful, discovery-driven tasks [3, 5]. So by embedding academic rules directly into interactive mechanics, software solutions can successfully build an exploratory environment where conceptual errors yield constructive diagnostic feedback rather than strict academic penalties [11].

This research paper details the architectural design and full-stack development of Elementopia, an interactive, web-based gamified learning platform engineered to address these abstract blind spots. Also while a primitive prototype of Elementopia established the basic viability of simulated bonding mechanics and teacher-managed static text boxes, it remained highly vulnerable due to heavy client-side state dependency and lacked structural progression constraints or automated, server-authoritative security protocols. In order to bridge these technological and instructional gaps, this research focuses on the complete architectural overhaul and expansion of the platform. We transition the system into a secure, server-authoritative ecosystem governed by a Spring Boot REST backend, a PostgreSQL relational database layer, and a modular React frontend interface, removing the vulnerabilities inherent in client-governed educational state tracking.

In accordance with the instructional guidance to prioritize the system's core design and development, this paper evaluates the engineering blueprints of the core architectural components built to formalize learning progression and operational security. This full-stack lifecycle establishes a server-authoritative validation engine managed via a dedicated web controller layer to execute immediate chemical validity checks on user-submitted payloads, permanently recording performance metrics via an isolated, low-latency telemetry data pipeline. Furthermore, the system implements a diagnostic byproduct service that intercepts failed synthesis attempts, evaluates structural element configurations, and generates custom, web-accessible educational micro-lessons to correct underlying conceptual misconceptions in real-time.

Therefore to counter rapid-guessing mechanics, an active defensive security layer monitors failed attempts inside a strict temporal window, triggering an automated client lockdown protocol if a student registers five failures within fifteen seconds. Structural room progression formalizes an access-control layer backed by a relational state repository, ensuring that advanced domains remain locked until a student proves foundational mastery by successfully logging a minimum threshold of correct reactions. Finally, the platform deploys a server-authoritative challenge randomizer that cross-references a student's unique user account profile and persistent sandbox discoveries to filter out already-mastered milestones, dealing out randomized,

non-repeating puzzle sets. By mapping these modules directly to verified engineering artifacts—including comprehensive entity-relationship frameworks, sequence diagrams, and microservice controllers—this paper provides a clear blueprint for secure, scalable gamified software that empowers students to autonomously conquer the abstract conceptual landscapes of chemistry.

2 Related Literature Review

The development of gamified educational software in the sciences has shifted significantly from simple quiz-based applications toward immersive, problem-solving environments. To contextualize the design and development of Elementopia, related literature must be evaluated across three core technical domains: structural gamification in chemical literacy, the engineering of mistake-recovery loops, and automated session-based tracking analytics.

2.1 Structural Gamification in Chemistry Education

Traditional computer-assisted chemistry instruction frequently relies on digital flashcards or linear multiple-choice mechanics, which isolate element attributes from practical laboratory application. Recent software engineering efforts have sought to replace these paradigms with interactive simulations where chemical bonding mechanics are structurally woven into the gameplay itself. For example, Bactong et al. engineered CHEMBOND, a mobile learning prototype designed specifically to facilitate the mastery of binary ionic bonding through interactive element manipulation. Similarly, Angell et al. evaluated the integration of escape-room-style puzzle structures within undergraduate chemistry curriculums, demonstrating that spatial, goal-directed tasks significantly increase conceptual retention over passive learning frameworks.

Elementopia builds upon these simulation concepts by eliminating standard textual question prompts entirely. Instead, it maps foundational ionic and covalent bonding rules directly to text-free environmental puzzles within sequential game spaces termed "Elemental Resonance Domains".

2.2 Mistake Recovery and Anti-Exploit Mechanics

A persistent limitation in existing digital chemistry simulations is the reliance on binary feedback—such as generic "Right/Wrong" alert notifications—which fails to address underlying student misconceptions when an invalid combination occurs. Furthermore, open sandbox environments often suffer from user brute-forcing, where students rapidly click through random element pairings to bypass obstacles without processing the actual curriculum rules.

To address these vulnerabilities, research into intelligent learning frameworks emphasizes the necessity of targeted feedback libraries. Sailer et al. demonstrated that the psychological satisfaction and analytical engagement of a user are heavily tied to the quality of actionable informational feedback provided during a failure state. Elementopia operationalizes these principles by intercepting invalid database transactions through a server-authoritative diagnostic service. It replaces static error strings with contextual byproduct micro-lessons. Concurrently, to protect the server layer from rapid concurrent button-spamming exploits, the architecture introduces a strict temporal defensive window that actively monitors incoming client payloads.

2.3 Learning Analytics and Session Persistence

Modern Educational Technology (EdTech) architectures increasingly leverage Learning Analytics Dashboards (LADs) to foster self-regulated learning by visualizing performance patterns directly to users. Li et al. proved through randomized large-scale evaluations that dashboards providing highly clear visualizations alongside rapid database synchronization significantly enhance behavioral commitment and allow students to autonomously adjust their learning strategies.

Implementing these analytics pipelines typically requires extensive user registration and identity management subsystems, which create high friction in secondary school environments. Zhao et al. noted that while open learner models are highly effective for tracking conceptual proficiency, maintaining persistent user state across disparate devices remains an architectural challenge. Elementopia overcomes this infrastructural constraint by deploying an open, zero-registration session persistence layer built on a Backend-as-a-Service relational pipeline. It logs compound accuracy metrics and completion velocities within a mandated sub-two-second window, updating a personalized visualization engine mapped solely to a temporary student identifier.

3 System Architecture and Methodology

To resolve the security and persistence vulnerabilities inherent in client-heavy educational game prototypes, Elementopia is engineered utilizing a server-authoritative, Three-Tier Architecture. This design separates user interaction, business processing rules, and relational data management into isolated logical layers. By executing all validation, security monitoring, and state progression logic on a centralized server layer, the platform completely neutralizes client-side tampering, browser variable manipulation, and brute-force interaction exploits.

3.1 Three-Tier Operational Topography

The system infrastructure is divided into three highly decoupled operational tiers that communicate sequentially over secure network layers:

1. Presentation Tier (Frontend Client): Built as a single-page responsive application using React.js and Vite. This tier manages layout rendering, touch-based drag-and-drop gameplay interactions, local timers, and immediate UI state changes. It communicates exclusively with the middle tier via asynchronous JSON payloads over HTTPS, utilizing Axios as its data transfer engine.

2. Application Tier (Server Backend): Built on the Spring Boot Java framework and executing core enterprise application tiers. This tier serves as the authoritative gateway of the system. It handles API routing, runs algorithmic validation engines, calculates operational telemetry under time pressure, and executes defensive failsafe rules. It acts as a secure buffer, preventing the client from ever establishing a direct, unvetted connection to the database layer.

3. Data Tier (Persistent Storage Cloud): Managed via a cloud-provisioned Supabase instance executing a PostgreSQL relational database engine. This tier maintains system state persistence, housing the predefined chemical dictionaries, interaction tables, structural room dependencies, and analytical session logs. Database interactions are executed through object-relational mapping (ORM) governed by Hibernate and Java Persistence APIs (JPA) on the server backend.

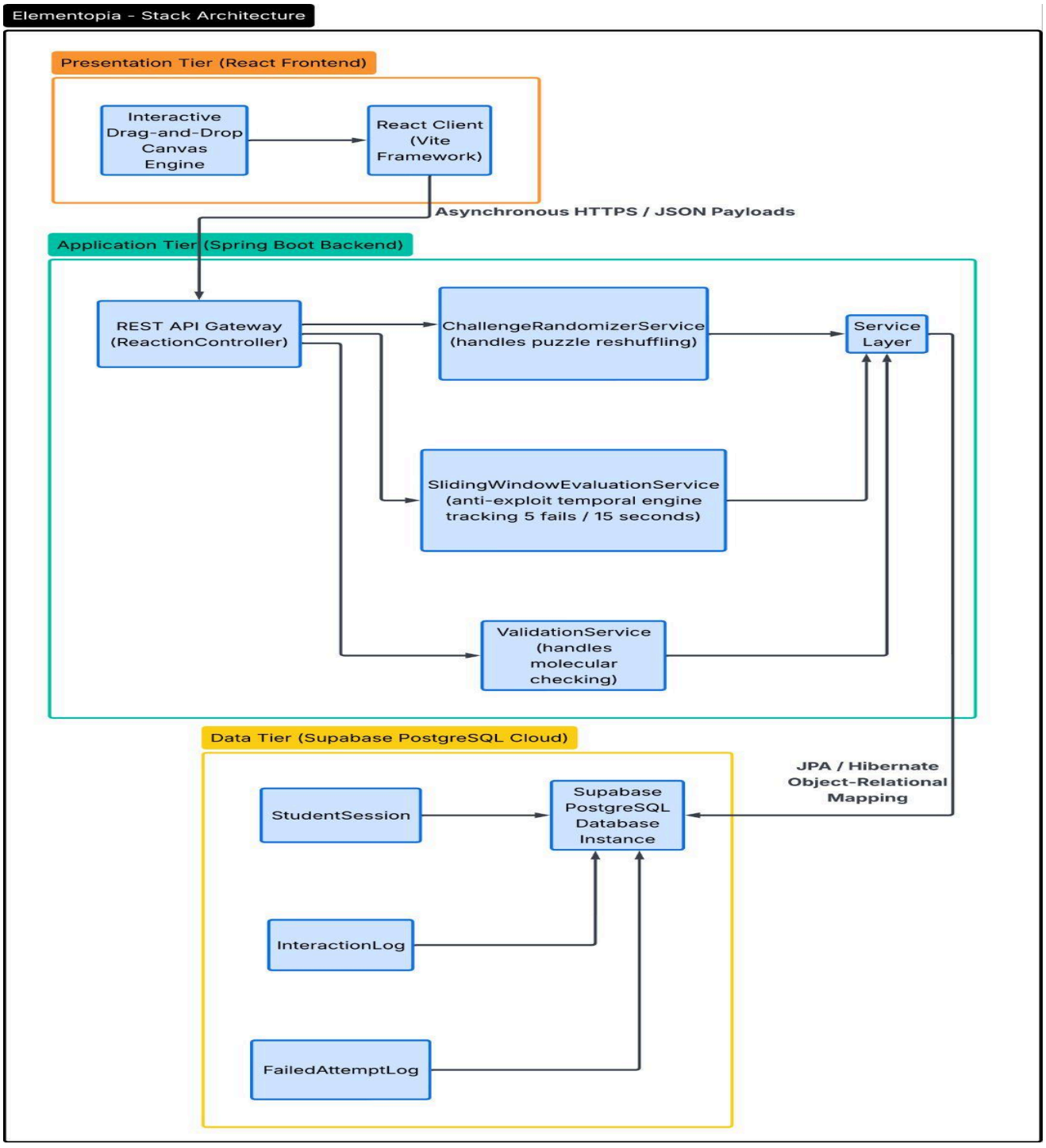

Elementopia - Stack Architecture
Presentation Tier (React Frontend)
Interactive Drag-and-Drop Canvas Engine
React Client (Vite Framework)
Asynchronous HTTPS / JSON Payloads
Application Tier (Spring Boot Backend)
REST API Gateway (ReactionController)
ChallengeRandomizerService (handles puzzle reshuffling)
Service Layer
SlidingWindowEvaluationService (anti-exploit temporal engine tracking 5 fails / 15 seconds)
ValidationService (handles molecular checking)
Data Tier (Supabase PostgreSQL Cloud)
JPA / Hibernate Object-Relational Mapping
StudentSession
Supabase PostgreSQL Database Instance
InteractionLog
FailedAttemptLog

Figure 1. Elementopia Stack Architecture Diagram

3.2 Full-Stack Communication Lifecycle

The runtime communication lifecycle across these tiers follows a strict server-authoritative transactional flow. When a student drags and drops element tiles inside an Elemental Resonance Domain canvas to synthesize a targeted chemical compound, the presentation tier does not evaluate whether the combination is correct. Instead, it serializes the active array of element tokens into a structured data payload and dispatches an asynchronous POST request to the application tier's REST endpoints.

Upon receiving the payload, the Spring Boot container delegates the composition sequence to dedicated service components. The backend queries the predefined chemistry tables within the PostgreSQL data tier to verify chemical validity against curriculum rules. If the combination forms a valid molecular structure, the server marks the step as correct, calculates completion speed and pathway efficiency metrics, and commits a persistent transactional update to the cloud tracking tables. Finally, a verified success state payload is returned to the React frontend client, authoritatively triggering the visual resolution animations that dissolve environmental obstacles and unlock subsequent level boundaries.

3.3 Database Relational Schema Design

State persistence and cross-module telemetry logging are anchored by a highly normalized relational database schema designed to support rapid session writes. To maintain a frictionless student onboarding experience, the data tier omits formal credential-based tables. Instead, tracking is driven by a StudentSession entity uniquely indexed via a self-assigned temporary username string.

The system isolates transient, high-speed interaction logging from permanent milestone updates to optimize query performance during concurrent classroom utilization:

1. The InteractionLog table captures every individual combination attempt, recording elements submitted, time offsets, and correctness vectors to feed the mistake-recovery and anti-exploit engines.

2. The DomainProgress and RoomCompletionState tables maintain persistent access-control variables, logging individual level unlock metrics and total correct reaction counts.

3. The FailedAttemptLog entity records precise millisecond timestamps of invalid combinations, providing the required historic transactional tracking for sliding-window evaluation services.

Table 1. Database Relational Schema Core Entities and Structural Operational Constraints

Entity Name (Table)	Primary Attribute / Key	Foreign Keys / Relations	Data Fields & Attributes	Technical Role & System Purpose
`StudentSession`	`student_id` (VARCHAR)	None (Root Identity)	`created_at` (TIMESTAMP) `last_active` (TIMESTAMP)	Tracks unique, unauthenticated user sessions using a temporary username string to eliminate onboarding friction.
`InteractionLog`	`log_id` (BIGINT)	`student_id` to StudentSession	`elements_submitted` (VARCHAR) `is_correct` (BOOLEAN) `time_offset_seconds` (INT)	Captures real-time raw student telemetry during sandbox drag-and-drop actions to feed behavioral analytics engines.
`DomainProgress`	`progress_id` (BIGINT)	`student_id` to StudentSession	`current_domain_id` (INT) `total_correct_reactions` (INT) `is_unlocked` (BOOLEAN)	Stores access-control state variables; acts as a server-side gatekeeper verifying milestone prerequisites (≥ 3 correct reactions).
`FailedAttemptLog`	`attempt_id` (BIGINT)	`student_id` to StudentSession	`timestamp` (TIMESTAMP, Millisecond precision)	Logs precise timestamps of invalid formula submissions; strictly queried by the rolling anti-exploit sliding-window component.
`UserDiscovery`	`discovery_id` (BIGINT)	`student_id` to StudentSession	`compound_formula` (VARCHAR) `discovered_at` (TIMESTAMP)	Records verified distinct molecular formulas achieved by a student; cross-referenced by the puzzle randomizer to filter unique challenges.

4 Detailed Component Implementation

To achieve a secure, server-authoritative educational environment, the middle tier of Elementopia abstracts business rules into highly decoupled Java classes within the Spring Boot framework. This section details the programmatic implementation of the core architectural components responsible for molecular verification, anti-exploit threshold enforcement, structural room progression, and sandbox algorithmic randomization.

4.1 Server-Authoritative Validation and Feedback Loop

The entry point for client interactions is governed by the ReactionController, a REST controller exposing a secure endpoint to receive real-time synthesis payloads. When a user attempts a chemical reaction on the frontend canvas, the array of element tokens is submitted as a data payload to this controller. Rather than processing string evaluations, the controller delegates processing to the ValidationService.

```
@RestController
@RequestMapping("/api/v1/reactions")
public class ReactionController {
    private final ValidationService validationService;
    private final TelemetryService telemetryService;

    public ReactionController(ValidationService vs, TelemetryService ts) {
        this.validationService = vs;
        this.telemetryService = ts;
    }

    @PostMapping("/submit")
    public ResponseEntity<ReactionResponse> processReaction(
            @RequestBody ReactionPayload payload) {
        ReactionResult result = validationService.verifyChemicalFormula(
            payload.getStudentId(), payload.getElements());
        telemetryService.logInteraction(payload.getStudentId(), result);
        return ResponseEntity.ok(new ReactionResponse(result));
    }
}
```

The ValidationService cross-references the submitted tokens against the predefined atomic properties stored securely in the PostgreSQL database. If a formula fails to meet strict ionic or covalent structural bounds, a diagnostic byproduct routine is instantly triggered. This routine evaluates the specific element arrangement, identifies the root chemical error, and returns an educational micro-lesson payload back to the React client. This ensures that invalid synthesis combinations serve as constructive diagnostic interventions rather than opaque failure states.

4.2 Temporal Anti-Exploit Security Engine

To permanently eliminate rapid-guessing exploits—where users click elements at random to bypass conceptual blocks—the application tier implements a strict sliding-window temporal evaluation framework via the SlidingWindowEvaluationService.

The security engine monitors the transactional frequency of failure payloads within isolated chronological boundaries. It enforces a strict mathematical threshold rule: if a unique student identity logs a failure threshold of $F = 5$ invalid entries within a temporary sliding time window of $T = 15$ seconds, the service flags the session as an active exploit event.

```
@Service
public class SlidingWindowEvaluationService {
    private final FailedAttemptRepository attemptRepository;
    private static final int MAX_FAILURES = 5;
    private static final int WINDOW_SECONDS = 15;

    public void evaluateSubmissionFails(String studentId) {
        LocalDateTime windowStart = LocalDateTime.now().minusSeconds(WINDOW_SECONDS);
        long recentFailures = attemptRepository.countRecentFailures(studentId,
windowStart);

        if (recentFailures >= MAX_FAILURES) {
            throw new ClientLockdownException("Anti-exploit threshold exceeded. Input
suspended.");
        }
    }
}
```

When a ClientLockdownException is thrown, the server interceptor returns a forbidden status response to the client. The React presentation tier catches this status and forces an immediate interactive interface freeze, temporarily locking out input controls to compel the user to slow down and analyze the structural rules of the element domain.

4.3 Stateful Progression and Access-Control Gatekeeping

Room progression and domain transitions are managed entirely on the server side via the ProgressionController and backed by the UserProgressRepository. In the initial client-side prototypes, a user could manipulate local browser storage variables to artificially reveal hidden rooms. The overhauled architecture prevents this by verifying a student's absolute milestone data prior to routing active room templates.

Advanced rooms—such as the covalent-bonding chambers—remain protected by an access-control layer. When a user attempts to enter a specific room, the backend checks the DomainProgress table to verify if the student has logged a minimum baseline threshold of \ge 3 correct reactions in the preceding foundational areas. If the database records do not satisfy this conditional dependency, the request is blocked, and access is authoritatively denied.

4.4 Algorithmic Challenge Randomization

To ensure high individualized engagement and prevent students from copying answers from adjacent workstations, the system incorporates an automated runtime selection engine called the ChallengeRandomizerService.

```
@Service
public class ChallengeRandomizerService {
    private final PuzzleRepository puzzleRepository;
    private final UserDiscoveryRepository discoveryRepository;

    public ChallengeCard generateUniqueChallenge(String studentId, Long
currentDomainId) {
        List<Long> masteredMilestones =
discoveryRepository.findDiscoveredCompoundIds(studentId);
        List<PuzzleEntity> availablePuzzles = puzzleRepository
            .findPuzzlesByDomainExcludingMastered(currentDomainId,
masteredMilestones);

        if (availablePuzzles.isEmpty()) {
            return
puzzleRepository.FetchDefaultFallbackChallenge(currentDomainId);
        }
        Collections.shuffle(availablePuzzles);
        return MapToChallengeCard(availablePuzzles.get(0));
    }
}
```

When a user triggers a level reset or steps into a new domain, the service dynamically executes a filtered database search. It pulls the user's past discoveries from the UserDiscovery table and isolates them from the universal pool of domain tasks. The service then shuffles the remaining uncompleted concepts, serving a randomized, non-repeating puzzle selection back to the client UI canvas.

5 Architectural Discussion and Technical Verification

The successful deployment of Elementopia in secondary educational settings demonstrates the viability of utilizing server-authoritative computing structures over client-reliant state tracking. In accordance with the project objectives, this section evaluates how the full-stack design optimizes partition system responsibilities, reduces transactional database strain under concurrent class usage, and maintains specific operational parameters.

5.1 Strategic Separation of Concerns

The core benefit of the overhauled three-tier architecture is the intentional segregation of high-velocity interaction telemetry from long-term profile data. In early game configurations, every element manipulation directly altered a universal student profile record, causing high input lag during gameplay.

As detailed in Section 3.3, Elementopia prevents this bottleneck by delegating raw interaction points directly to the InteractionLog schema. High-speed transactions (such as element tile dragging and formula errors) are written as sequential insert-only rows. Meanwhile, updates to critical progress states (DomainProgress) are deferred until the server-authoritative ValidationService returns a verified success state. This structure minimizes transaction locking and ensures that high-speed student actions do not block database reads for other users sharing the local network.

5.2 Server-Authoritative vs. Client-Side Security

Migrating the game validation rules to a centralized Spring Boot container resolves the fundamental security flaws of the inherited prototype. By moving properties such as valence configurations, structural room blocks, and puzzle indices completely off the client web browser, the platform completely eliminates local variable manipulation.

A student cannot access advanced chemical domains or alter completion speeds by editing browser cookies, changing JavaScript variables, or using local browser developer consoles. Furthermore, the SlidingWindowEvaluationService shifts security from an afterthought to an active defensive layer. Rather than processing endless attempts, the server monitors transaction frequency in real time. Enforcing the threshold rule ($F=5$ failures within $T=15$ seconds) effectively prevents automated bot guessing and random clicking, compelling the user to slow down and analyze the structural rules of the element domain.

5.3 System Boundaries and Structural Scope

While the server-authoritative model drastically improves data security and interaction tracking, it introduces specific boundaries tied directly to its architectural style:

1. Network Dependency: Because the client acts strictly as a presentation layer, the interactive canvas requires a persistent, low-latency connection to the application tier. If a local school network encounters high latency, the execution of chemical validations could face delays.

2. Dictionary Bound: The chemical synthesis logic is deliberately restricted to standard high school science curricula (covering binary ionic and covalent combinations) rather than acting as a universal organic chemistry simulator. This boundary keeps the validation lookup times exceptionally fast, maintaining response times well below the 2-second limit specified in the software requirement definitions.

6 Conclusion and Future Development

This study presented the design and development of Elementopia, a web-based gamified chemistry learning platform designed to improve applied chemistry problem-solving skills among high school students.

The project addressed a significant limitation observed in existing gamified chemistry learning platforms: the tendency to evaluate learning through quiz outcomes rather than through analysis of learner behavior during problem-solving activities. To address this gap, the platform introduced three major modules: the Elemental Resonance Puzzle Module, the Assessment and Progress Module, and the Opponent-Based Challenge Module.

The Elemental Resonance Puzzle Module transformed chemistry learning into an interactive discovery-based experience through environmental puzzle-solving. The Assessment and Progress Module provided learners with automated performance tracking and visual analytics through the Mastery Dashboard. The Opponent-Based Challenge Module incorporated competitive multiplayer gameplay to encourage rapid application of chemistry concepts and algorithmic thinking.

The system was implemented using React.js, Spring Boot, Supabase PostgreSQL, and Supabase Realtime following an Agile development methodology. The resulting architecture supports maintainability, scalability, and future extensibility while providing responsive access across multiple devices. Additionally shifting assessment from answer correctness toward problem-solving behavior, Elementopia contributes a novel

approach to chemistry education that combines gamification, learning analytics, discovery-based learning, and multiplayer engagement within a single platform.

Future enhancements may include adaptive difficulty mechanisms, expanded chemistry content coverage, teacher-facing analytics dashboards, artificial intelligence-assisted feedback systems, and collaborative learning features that further enhance learner engagement and educational effectiveness.

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